

Student Project: Mini-HPC and Hybrid HPC-Big Data Clusters

Objective

By the end of this project, students will:

- Understand fundamentals of High-Performance Computing (HPC) and Big Data processing.
- Set up a virtual 3-node cluster (1 master + 2 workers) using VirtualBox.
- Implement distributed ML tasks using MPI and mpi4py (Task 1).
- Build a Docker Swarm cluster to run distributed Spark jobs (Task 2).
- Run and evaluate distributed models on a real bioinformatics gene expression dataset.
- Optionally manage jobs using SLURM scheduler for Task 1.

Project Timeline (4 Weeks Suggested)

Week 1: Set up virtual machines and configure networking (Task 1 base)

Week 2: Install MPI, Python, and dependencies; complete distributed ML (Task 1)

Week 3: Initialize Docker Swarm; Deploy Spark Cluster (Task 2 setup)

Week 4: Run distributed bioinformatics ML with Spark; generate final report

Deliverables

1. Working Mini-HPC Cluster (3 VMs) with MPI + Passwordless SSH
2. Distributed ML Task using Digits Dataset with OpenMPI
3. Distributed Bioinformatics Analysis with OpenMPI (Task 1)
4. Docker Swarm Spark Cluster running PySpark ML on gene expression data (Task 2)
5. Documentation Folder (README.md, hostfile, Docker YAMLs, scripts, screenshots, reports)

Project Tasks

Task 1: Mini-HPC Cluster (Traditional)

Task 2: Hybrid HPC + Big Data Cluster

Guidelines

- Work in teams of 2–3 students.
- Maintain a detailed setup log.
- Comment all scripts and document any issues.
- Take screenshots at key steps (SSH setup, MPI tests, Spark UI).
- Submit final documentation and source code.

Evaluation Criteria

VM setup and network configuration (Task 1): 20 pts

Successful MPI Distributed ML (Task 1): 20 pts

Bioinformatics ML pipeline on MPI (Task 1): 20 pts

Docker Swarm Cluster setup (Task 2): 15 pts

Distributed Spark ML execution (Task 2): 15 pts

Report & presentation (combined): 10 pts

Bonus Extensions

- Add a 4th VM node and benchmark performance gains.
- Deploy HDFS alongside Spark and store datasets on HDFS.
- Integrate TensorFlow Distributed Training with MPI or Spark.
- Use JupyterHub deployed on the cluster for interactive work.
- Monitor containers and services using Grafana + Prometheus.

Submission Format

project_hpc_hybrid_cluster/

├── README.md

├── hostfile (Task 1)

├── distributed_mnist.py (Task 1)

```
└─ distributed_gene_analysis.py (Task 1)
└─ spark-stack.yml (Task 2)
└─ distributed_gene_expression_analysis.py (Task 2)
└─ bioinfo_data/
    └─ leukemia_expression.csv
└─ screenshots/
└─ Final_Report.pdf
└─ ml_job.slurm (Optional SLURM Task 1)
```

Congratulations

Upon successful completion, you will have:

- Built both a Mini-HPC cluster and a Big Data hybrid cluster.
- Run real distributed machine learning jobs across your nodes.
- Worked with real-world bioinformatics data and Spark MLlib.